
*Responsible AI-Assisted Early Pediatric Vision Risk Detection
Before Age 3*

*A Hospital-Integrated Framework for Early Identification and
Referral Support for Strabismus and Amblyopia*

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Disclaimer

*This document presents a conceptual research framework intended for
academic discussion and future research exploration. It does not describe a
validated clinical system, diagnostic tool, or medical device, and should not be
interpreted as medical advice or clinical guidance.*

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1. Abstract

Early childhood vision disorders such as amblyopia and strabismus remain significant and largely preventable causes of long-term visual impairment, collectively affecting an estimated 100 million children globally. Treatment outcomes depend heavily on early identification, particularly during the first years of life, when the visual system undergoes critical neurodevelopment. However, current pediatric vision screening programs are often implemented at preschool or school-entry age, which may be beyond the most sensitive period for optimal intervention (American Association for Pediatric Ophthalmology and Strabismus, n.d.; Hashemi et al., 2018, 2019).

In addition to timing limitations, existing care pathways are frequently fragmented across pediatric screening, referral processes, and specialist evaluations, resulting in diagnostic delays and incomplete follow-up. While artificial intelligence has demonstrated promise in pediatric vision analysis, much of the existing research focuses on model-level diagnostic accuracy rather than real-world clinical integration and healthcare workflow implementation (He et al., 2025; Mao et al., 2021; Murali et al., 2021; Shu et al., 2024).

This white paper presents a conceptual framework for responsible AI-assisted pediatric vision risk identification within routine healthcare systems, with a specific focus on children under the age of three. The proposed approach emphasizes integration into pediatric care visits, the use of electronic health record (EHR) signals, clinician-in-the-loop decision support, and structured referral pathways to facilitate timely ophthalmologic evaluation.

Rather than proposing an autonomous diagnostic system, this work highlights the importance of aligning AI-based risk assessment with healthcare system workflows to address gaps in care continuity, adherence to follow-up, and opportunities for early intervention (World Health Organization, 2023).

The framework aims to contribute to future research directions in pediatric vision care, responsible AI deployment, and healthcare system integration, with the ultimate goal of reducing preventable childhood vision loss through earlier and more equitable access to specialist evaluation.

2. Introduction

2.1 Background

Why Pediatric Vision Matters

Pediatric vision health is fundamental to healthy childhood development. During the first years of life, visual function plays a critical role in the development of cognitive, motor, language, and social skills. Disruptions to normal visual development during this sensitive period can have lasting consequences that extend beyond vision alone, influencing educational achievement, quality of life, and long-term functional outcomes (National Eye Institute, n.d.-b).

Among the most common pediatric vision disorders are amblyopia (commonly known as *lazy eye*) and strabismus (ocular misalignment). Amblyopia is recognized as one of the leading causes of preventable visual impairment in children, while strabismus affects an estimated 2–4% of the pediatric population worldwide. Together, these conditions represent a significant public health concern, with estimates suggesting that approximately 100 million children globally may be affected by amblyopia (Hashemi et al., 2018, 2019; National Eye Institute, n.d.-a).

Furthermore, because these conditions are largely invisible to parents and caregivers in early childhood, young children cannot report visual suppression, intermittent diplopia, or reduced acuity in one eye; the burden of identification falls entirely on healthcare systems and screening programs (American Association For Pediatric Ophthalmology And Strabismus, n.d.).

Unlike many adult eye diseases, the effectiveness of treatment for amblyopia and strabismus depends heavily on early recognition during the period of visual system development. Delayed identification may result in reduced visual acuity, impaired binocular vision, loss of stereopsis (depth perception), and, in some cases, irreversible visual deficits despite later treatment (National Eye Institute, n.d.-b).

Consequently, pediatric vision health should be viewed not only as an ophthalmologic concern but also as a broader public health priority. Improving the timeliness of identifying children at risk can reduce preventable vision loss and improve developmental outcomes across the lifespan.

2.2 Importance of Early Detection Before Age 3

Why timing matters

Early childhood is a critical period for visual system development, during which neural pathways responsible for binocular vision, visual acuity, and depth perception undergo rapid, time-sensitive maturation. This developmental window, often referred to as the critical or sensitive period of visual development, is characterized by high neuroplasticity, during which the visual cortex remains highly responsive to sensory input. After approximately age 7-8, this plasticity diminishes significantly, reducing the effectiveness of amblyopia treatment (National Eye Institute, n.d.-b).

Clinical evidence in pediatric ophthalmology has consistently demonstrated that the timing of intervention significantly affects treatment outcomes for conditions such as amblyopia and strabismus. When these conditions are identified and treated in early childhood, particularly during the early developmental years, there is a substantially higher likelihood of achieving favorable visual outcomes than with later intervention. Delayed detection, by contrast, may result in persistent deficits in visual acuity, impaired binocular coordination, and reduced stereopsis, some of which may become only partially reversible even with treatment (Hashemi et al., 2018, 2019).

Despite the clinical importance of early intervention, many existing screening programs are conducted at preschool or school-entry age, which may occur after the most sensitive period of visual development has already passed. In contrast, children under the age of three frequently interact with healthcare systems through routine well-child visits and developmental assessments, representing a critical yet underutilized opportunity for earlier identification of potential vision concerns (American Academy of Ophthalmology, n.d.; American Association For Pediatric Ophthalmology And Strabismus, n.d.).

However, identifying risk at this stage presents challenges, as clinical signs may be subtle, intermittent, or difficult to assess consistently in very young children. Additionally, variability in screening practices, referral pathways, and follow-up processes can further delay specialist evaluation even when concerns are initially identified.

These considerations highlight that improving outcomes in pediatric vision care requires not only effective screening strategies, but also a deeper understanding of how early signals can be integrated into healthcare workflows to support timely and appropriate clinical evaluation during the most developmentally sensitive period.

2.3 Problem Statement

Despite the existence of established pediatric vision screening programs and increasing awareness of early childhood eye health, significant gaps remain in the timely identification and management of vision-related conditions such as amblyopia and strabismus. These gaps are not solely related to diagnostic capability but are largely rooted in system-level, workflow, and care continuity challenges (American Association For Pediatric Ophthalmology And Strabismus, n.d.; Park et al., 2025).

One of the primary limitations is that current screening approaches are often implemented at preschool or school-entry stages, which may already fall outside the most sensitive period for optimal visual development. As a result, children who exhibit early risk indicators prior to formal screening age may not be identified in time for the most effective intervention window (American Academy of Ophthalmology, n.d.; American Association For Pediatric Ophthalmology And Strabismus, n.d.).

In addition, even when potential vision concerns are identified during routine pediatric visits or screening programs, the pathway from initial detection to definitive ophthalmologic evaluation is often fragmented. Referral processes may be inconsistent, follow-up completion rates may be low, and communication between pediatric care providers and specialist services is not always seamless. These discontinuities can lead to delays in diagnosis and intervention, even when initial concern has been appropriately raised (American Association For Pediatric Ophthalmology And Strabismus, n.d.; Park et al., 2025).

Another important limitation is that current systems tend to operate reactively rather than proactively. Screening is typically episodic rather than continuously integrated into routine pediatric care workflows. This limits the ability to identify evolving or subtle risk patterns over time, particularly in very young children, where clinical signs may not be consistently observable during a single visit (American Association For Pediatric Ophthalmology And Strabismus, n.d.; Park et al., 2025).

While artificial intelligence and data-driven tools have been increasingly explored in medical imaging and diagnostic contexts, much of the existing work has focused on improving classification performance rather than addressing real-world implementation challenges such as care coordination, workflow integration, and longitudinal follow-up. As a result, there remains a gap between technical capability and practical clinical impact (Liew & Agaian, 2025; Yanovitch, 2024).

Perhaps most critically, current pediatric vision AI research is primarily framed around model-level performance evaluation, particularly image-based classification tasks such as detecting strabismus or amblyopia from photographs. While these approaches demonstrate technical progress in algorithmic accuracy, they do not fully address the broader clinical question of how such systems integrate into real-world healthcare delivery (He et al., 2025; Mao et al., 2021; Murali et al., 2021; Shu et al., 2024).

From a systems perspective, the more clinically meaningful question extends beyond whether an AI model can detect a condition to whether AI-enabled tools can support healthcare systems in ensuring that children identified as potentially at risk are successfully guided through timely referral and ophthalmologic evaluation.

This distinction between algorithmic detection performance and end-to-end clinical pathway integration represents a fundamental gap in the current literature. It highlights the need to move beyond isolated model evaluation to frameworks that integrate workflows, coordinate care, and support real-world implementation within pediatric healthcare systems (Park et al., 2025; Yanovitch, 2024).

This white paper is positioned to address this gap by focusing on early risk identification embedded within routine pediatric care workflows, rather than standalone diagnostic prediction systems.

2.4 Purpose and Scope of This White Paper

The purpose of this white paper is to present a conceptual research framework exploring how responsible artificial intelligence may support earlier identification of pediatric vision risk within routine healthcare systems, particularly for children under the age of three. The focus of this work is not on developing an autonomous diagnostic system, but rather on examining how AI-

assisted decision support, clinical workflows, and healthcare system integration may collectively contribute to earlier recognition and timelier referral of children who may require ophthalmologic evaluation.

This work synthesizes findings from existing literature in pediatric vision screening, healthcare AI, and clinical workflow research, and identifies key gaps between technical advancements and real-world clinical implementation. In particular, it highlights the importance of understanding not only detection performance, but also system-level factors such as referral pathways, follow-up adherence, and care coordination between pediatric and ophthalmology services (American Academy of Ophthalmology, n.d.; World Health Organization, 2021; Zheng et al., 2021).

The proposed framework is intended to serve as a conceptual foundation for future research and interdisciplinary collaboration among clinicians, researchers, healthcare administrators, and artificial intelligence practitioners. It is not intended to provide clinical recommendations, validated algorithms, or guidance for immediate clinical deployment.

Rather, this white paper aims to contribute to ongoing discussions about how healthcare systems can evolve to better integrate early risk identification into routine pediatric care, with a strong emphasis on responsible AI principles, patient safety, and real-world feasibility.

3. Literature Review Summary

3.1 Overview of Existing Studies

A focused review informs this white paper, drawing on peer-reviewed literature examining pediatric vision screening, amblyopia, strabismus, artificial intelligence applications in ophthalmology, responsible AI, and healthcare implementation. The review included systematic reviews, meta-analyses, validation studies, population-based screening investigations, bibliometric analyses, and expert commentaries published between 2018 and 2025. Rather than conducting a formal systematic review, the objective was to synthesize current evidence relevant to the proposed conceptual framework and to identify gaps between technical advances in artificial intelligence and their integration into routine pediatric healthcare workflows (Hashemi et al., 2018, 2019; World Health Organization, 2023).

Over the past decade, research in pediatric vision screening has expanded considerably, with growing interest in the application of artificial intelligence to support the early identification of conditions such as amblyopia and strabismus. Existing studies have explored a wide range of approaches, including traditional screening methods, smartphone-based image analysis, automated image classification, deep learning algorithms, and AI-assisted diagnostic support systems (He et al., 2025; Mao et al., 2021; Murali et al., 2021; Shu et al., 2024).

Collectively, the literature indicates that artificial intelligence has the potential to improve the accuracy, efficiency, and accessibility of pediatric vision screening. Several studies have reported

promising diagnostic performance using facial photographs, ocular alignment analysis, retinal imaging, and digital screening devices. More recently, novel approaches have explored fixational eye movement analysis as a potential direct biomarker for amblyopia detection, demonstrating promising results in distinguishing amblyopic from normal visual function through AI-assisted eye tracking (Rakshit et al., 2026).

In parallel, public health and clinical research continues to emphasize the importance of early childhood vision screening and timely intervention. Population-based screening investigations have further highlighted the burden of undetected pediatric vision disorders at the community level. A nationwide school vision screening survey conducted in Armenia identified vision impairment in approximately 9.7% of school-aged children, with strabismus accounting for 3.8% of diagnoses, underscoring the importance of systematic early screening programs and the potential public health impact of earlier identification (Jrbashyan et al., 2022).

Multiple studies have demonstrated that treatment outcomes for amblyopia and strabismus are generally more favorable when these conditions are identified during early developmental stages, highlighting the importance of minimizing delays between risk identification and definitive ophthalmologic evaluation (American Academy of Ophthalmology, n.d.; National Eye Institute, n.d.-b).

These findings suggest that AI-assisted approaches may complement existing clinical practice by supporting earlier identification of children who require specialist evaluation (He et al., 2025; Mao et al., 2021; Murali et al., 2021; Shu et al., 2024).

More recently, researchers have also begun to emphasize responsible AI principles in healthcare, including fairness, transparency, explainability, privacy protection, and clinician oversight in the development of AI-based clinical decision support systems. These considerations are particularly important in pediatric healthcare, where patient safety, ethical responsibility, and family-centered care remain central to clinical decision-making (U.S. Food and Drug Administration, n.d.; World Health Organization, 2023).

Although these studies collectively demonstrate substantial progress in pediatric vision AI and early childhood screening, they also highlight important gaps in translation to clinical practice. In particular, there remains limited emphasis on integrating AI systems into real-world healthcare workflows, ensuring continuity of care, and bridging the gap between early risk identification and successful ophthalmologic evaluation (U.S. Food and Drug Administration, n.d.; World Health Organization, 2023).

3.2 Identified Gaps in Current Research

Key Gaps Identified from the Literature

A review of the current literature indicates that, despite significant technological progress, several important research gaps remain before AI-assisted pediatric vision screening can be effectively integrated into routine healthcare practice (Mao et al., 2021; Shu et al., 2024; World Health Organization, 2023).

The first gap relates to the primary focus of existing studies. Most published research emphasizes algorithm development and diagnostic accuracy rather than examining how AI systems can be incorporated into real-world pediatric care pathways (He et al., 2025; Mao et al., 2021; Murali et al., 2021).

Second, relatively few studies consider the complete patient journey beyond initial screening. Challenges related to referral completion, follow-up adherence, continuity of care, and communication between pediatric providers and ophthalmology services remain insufficiently addressed (World Health Organization, 2023).

Third, although early childhood is recognized as the most critical period for visual development, comparatively limited research specifically focuses on AI-assisted risk identification for children under three years of age, where clinical assessment may be more challenging and opportunities for early intervention are greatest (American Academy of Ophthalmology, n.d.; National Eye Institute, n.d.-b).

Fourth, most published AI models are evaluated under controlled research conditions. There remains a need for prospective clinical validation, multi-site evaluation, and assessment within routine healthcare environments to better understand real-world performance and implementation feasibility (He et al., 2025; Shu et al., 2024).

Finally, while responsible AI has become an increasingly important topic in healthcare, relatively few pediatric vision studies comprehensively address issues such as fairness across diverse populations, explainability for clinicians, privacy protection, governance, and long-term safety monitoring throughout deployment. (World Health Organization, 2023)

Taken together, these findings suggest that future research should move beyond isolated model development toward integrated healthcare frameworks that combine responsible AI principles with clinical workflows, electronic health record integration, clinician oversight, and structured referral pathways. This broader systems perspective provides the conceptual foundation for the framework proposed in this white paper.

4. Why Hospital Integration?

4.1 Limitations of School-Based Screening

Why Existing Screening Approaches May Miss, the Earliest Window

School-based vision screening programs have played an important role in identifying vision problems among children and have contributed significantly to public health efforts worldwide. However, these programs are typically conducted during preschool or school-entry years, after many children have already passed the earliest and most neurodevelopmentally sensitive stages of visual development (American Academy of Ophthalmology, n.d.; National Eye Institute, n.d.-b).

For conditions such as amblyopia and strabismus, delays in identification may reduce the effectiveness of treatment and increase the risk of persistent visual impairment. Consequently, while school-based screening remains valuable, it may not provide the earliest opportunity for identifying children who could benefit from ophthalmologic evaluation (American Academy of Ophthalmology, n.d.; Hashemi et al., 2018).

School screening should therefore be viewed as one component of a broader pediatric vision care strategy rather than the sole point of identification.

4.2 Role of Pediatric Visits

Routine Pediatric Care as an Opportunity

Children under three years of age typically attend multiple well-child visits during early development. These encounters provide repeated opportunities for developmental assessment, growth monitoring, immunization, and caregiver education (AAP, n.d.; National Eye Institute, n.d.-a).

Unlike episodic school screening, pediatric visits occur throughout infancy and early childhood, allowing healthcare providers to observe developmental changes over time. This continuity creates opportunities to recognize subtle clinical indicators, document family history, assess developmental milestones, and identify emerging vision-related concerns before formal school screening occurs (American Academy of Ophthalmology, n.d.; American Association For Pediatric Ophthalmology And Strabismus, n.d.).

Although routine pediatric visits are not intended to replace comprehensive ophthalmologic examinations, they represent a potentially valuable setting for earlier risk identification and appropriate referral when supported by structured clinical workflows.

4.3 EHR and Workflow Opportunities

Leveraging Existing Healthcare Information

Modern healthcare systems routinely collect a wide range of clinical information through Electronic Health Records (EHRs). These records may include demographic information,

developmental assessments, birth history, family history, prior clinical observations, referral records, and longitudinal documentation across multiple pediatric visits (HL7 International, n.d.; Office of the National Coordinator for Health Information Technology, n.d.).

While these data are primarily collected to support clinical care, they may also provide opportunities for identifying patterns associated with increased vision risk when interpreted within appropriate clinical and ethical frameworks (NIST, 2023; World Health Organization, 2023).

Rather than relying on a single screening event, longitudinal information captured across routine pediatric care may support more comprehensive risk assessment over time. Integrating responsible AI-assisted decision support within existing healthcare workflows could help clinicians recognize children who may benefit from earlier ophthalmologic evaluation while maintaining clinician oversight and established standards of care.

5. Proposed Framework

5.1 System Overview

Overview of the Proposed Framework

This white paper proposes a conceptual hospital-integrated framework designed to support earlier identification of pediatric vision risk among children under the age of three through responsible artificial intelligence (AI)-assisted clinical decision support. The framework is intended to complement, rather than replace, existing pediatric healthcare practices by integrating AI within routine clinical workflows and maintaining clinician oversight throughout the decision-making process (NIST, 2023; World Health Organization, 2021, 2023).

The proposed framework recognizes that early identification is only one component of effective pediatric vision care. Equally important are timely referral, continuity of care, communication between healthcare providers, and appropriate follow-up to ensure that children identified as potentially at risk receive comprehensive ophthalmologic evaluation (American Academy of Ophthalmology, n.d.).

Rather than functioning as an autonomous diagnostic system, the proposed framework emphasizes collaboration between healthcare professionals and AI-assisted risk assessment. Information collected during routine pediatric encounters, together with relevant electronic health record (EHR) data, may support clinicians in recognizing children who could benefit from additional assessment while preserving clinical judgment as the primary basis for decision-making (NIST, 2023; World Health Organization, 2023).

The framework presented in this white paper is conceptual and intended to stimulate future interdisciplinary research involving pediatric medicine, ophthalmology, health informatics, implementation science, and responsible AI.

5.2 Workflow Architecture (Figure 1)

Figure 1

Responsible AI-Assisted Early Pediatric Vision Risk Detection Before Age 3: Hospital-Integrated Workflow Framework

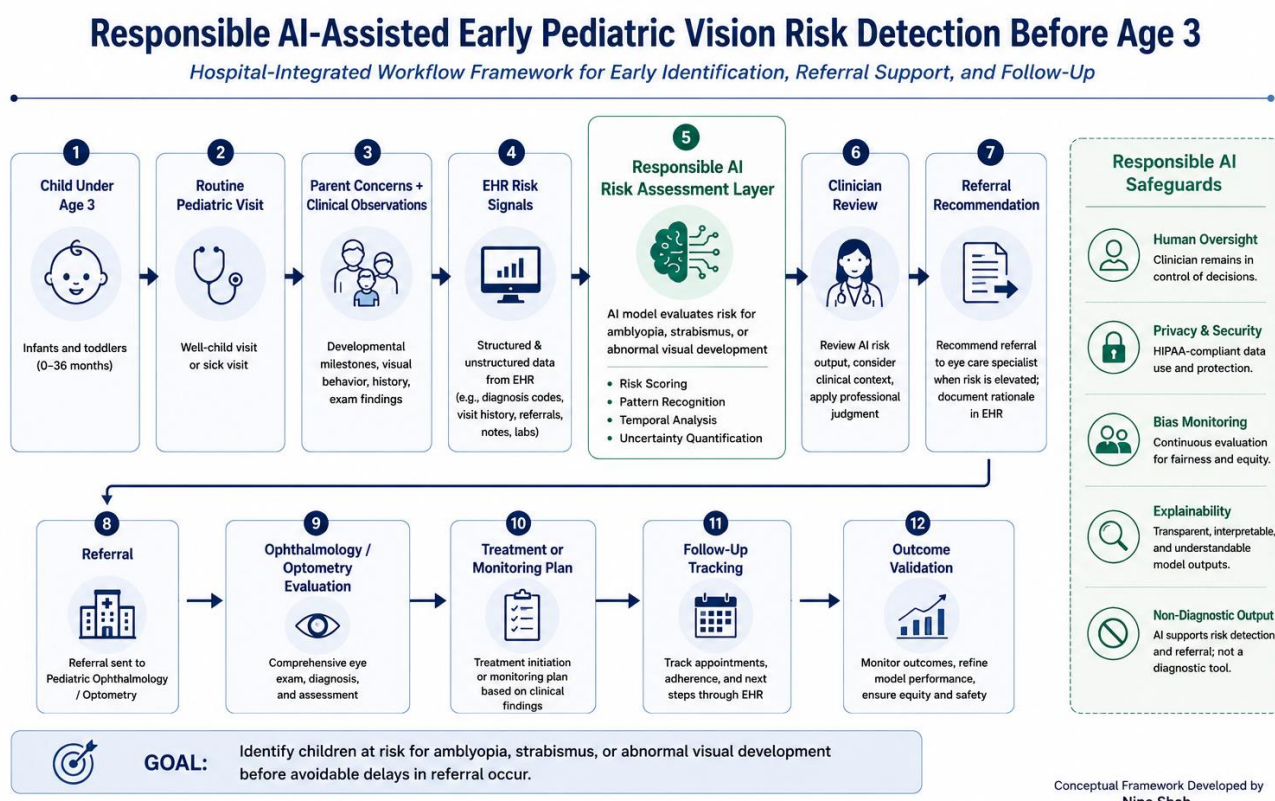


Figure 1. Responsible AI-Assisted Early Pediatric Vision Risk Detection Before Age 3: Hospital-Integrated Workflow Framework for Early Identification, Referral Support, and Follow-Up

Conceptual Framework Developed by
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Following the figure:

Figure 1 illustrates the conceptual workflow proposed in this white paper. The process begins during routine pediatric healthcare visits, where demographic information, developmental assessments, clinical observations, family history, and other relevant healthcare information are documented within the electronic health record. Rather than relying on a single screening event,

the framework considers longitudinal clinical information that accumulates across multiple pediatric encounters (NIST, 2023; World Health Organization, 2021, 2023).

A conceptual AI-assisted risk layer supports clinicians by identifying patterns that may indicate an increased likelihood of pediatric vision concerns requiring specialist evaluation. Importantly, AI-generated outputs are intended to function solely as clinical decision-support recommendations. Final assessment, referral decisions, and subsequent management remain under the supervision of qualified healthcare professionals (NIST, 2023; World Health Organization, 2023).

Children identified as potentially at increased risk may then be referred for comprehensive ophthalmologic evaluation, where formal diagnosis and treatment decisions are performed according to established clinical standards (American Academy of Ophthalmology, n.d.).

The framework therefore emphasizes continuity of care across the entire clinical pathway, from initial pediatric encounters through specialist evaluation.

5.3 AI Risk Layer (Conceptual Design)

AI as Clinical Decision Support

Within the proposed framework, the AI risk layer is conceptualized as a clinical decision-support component designed to assist healthcare professionals in recognizing children who may benefit from additional ophthalmologic assessment. The framework does not propose autonomous diagnosis or independent clinical decision-making by artificial intelligence (NIST, 2023; World Health Organization, 2023).

Conceptually, the AI risk layer may incorporate information routinely collected during pediatric care, including developmental assessments, documented clinical observations, family history, prior healthcare encounters, referral history, and other appropriate structured clinical information available within electronic health records. The specific data elements, analytical methods, and implementation strategies are intentionally left undefined, as they represent important areas for future investigation rather than established components of this framework.

Instead of generating definitive diagnoses, the AI system is envisioned as supporting risk stratification by highlighting children who may warrant closer clinical review. This approach aligns with responsible AI principles by emphasizing transparency, clinician oversight, and patient safety while minimizing unnecessary reliance on automated decision-making (NIST, 2023; U.S. Food and Drug Administration, n.d.; World Health Organization, 2023).

5.4 Clinician-in-the-Loop Design

Maintaining Human Oversight

A central principle of the proposed framework is that artificial intelligence should augment, rather than replace, clinical expertise. Throughout the proposed workflow, licensed healthcare professionals remain responsible for interpreting clinical findings, evaluating AI-assisted recommendations, determining the need for specialist referral, and making all final clinical decisions (NIST, 2023; World Health Organization, 2023).

Maintaining clinician involvement at every stage promotes patient safety, supports individualized clinical judgment, and reduces the risk of inappropriate reliance on algorithmic outputs. Human oversight also provides opportunities to consider contextual factors—including developmental history, caregiver concerns, physical examination findings, and other clinical information—that may not be fully represented within structured healthcare data (NIST, 2023; World Health Organization, 2023).

This clinician-in-the-loop approach further supports responsible AI by reinforcing accountability, transparency, and ethical decision-making within pediatric healthcare (NIST, 2023). By positioning AI as an assistive technology rather than an autonomous decision-maker, the proposed framework seeks to facilitate responsible integration of emerging technologies into existing clinical workflows while maintaining the highest standards of patient care.

6. Responsible AI Requirements

Artificial intelligence has the potential to support earlier identification of pediatric vision risk; however, its integration into clinical practice requires careful consideration of ethical, technical, and healthcare system responsibilities. Given the unique vulnerabilities of pediatric populations, AI systems intended for clinical decision support should be developed and evaluated according to principles that prioritize patient safety, fairness, transparency, and clinician oversight (NIST, 2023; World Health Organization, 2023).

The proposed framework presented in this white paper is grounded in responsible AI principles that seek to complement existing healthcare practices while maintaining trust between healthcare providers, patients, and families. Rather than emphasizing algorithmic performance alone, responsible AI requires consideration of how technology interacts with clinical workflows, healthcare equity, data governance, and long-term patient outcomes. (NIST, 2023; World Health Organization, 2023).

The following principles represent key requirements for future research and eventual clinical implementation.

6.1 Fairness and Bias

AI systems intended for pediatric healthcare should demonstrate equitable performance across diverse patient populations. Differences in ethnicity, skin tone, socioeconomic background, geographic location, and healthcare access may influence both the availability of clinical data and model performance. Future AI systems should therefore be evaluated for potential bias and should undergo validation across representative populations to reduce the risk of unequal healthcare outcomes (NIST, 2023; World Health Organization, 2023).

Ensuring fairness requires continuous monitoring throughout development and deployment rather than relying solely on initial model evaluation (NIST, 2023).

6.2 Privacy and Data Protection

Pediatric healthcare information represents highly sensitive personal data requiring appropriate safeguards throughout data collection, storage, analysis, and clinical use. Any future implementation of AI-assisted decision support should comply with applicable privacy regulations and institutional governance requirements while minimizing unnecessary collection or sharing of patient information (NIST, 2023; World Health Organization, 2021).

In the context of pediatric populations specifically, additional protections may be warranted given that patients are unable to provide independent informed consent. Parental or guardian consent processes should be clearly defined and accessible to families from diverse linguistic and cultural backgrounds before any AI-assisted risk assessment is conducted (AAP, n.d.; World Health Organization, 2021).

Whenever possible, healthcare organizations should adopt privacy-preserving approaches that support responsible use of clinical data while maintaining public trust and protecting children's rights (NIST, 2023; U.S. Food and Drug Administration, n.d.).

6.3 Explainability

Clinical decision support systems should provide information that is understandable and interpretable by healthcare professionals. AI-generated recommendations should assist clinicians by highlighting relevant findings or potential areas of concern rather than functioning as "black-box" predictions that cannot be meaningfully interpreted (NIST, 2023; U.S. Food and Drug Administration, n.d.; World Health Organization, 2023, 2023).

Explainable AI may improve clinician confidence, facilitate informed clinical decision-making, and support communication with caregivers regarding referral recommendations.(NIST, 2023; World Health Organization, 2023).

6.4 Safety and Clinical Boundaries

Patient safety should remain the highest priority throughout the development and use of AI-assisted clinical systems. The proposed framework does not envision AI as an autonomous diagnostic tool but rather as a mechanism for supporting clinician awareness and risk assessment within established healthcare workflows (NIST, 2023; U.S. Food and Drug Administration, n.d.; World Health Organization, 2023).

Any future clinical implementation should undergo rigorous technical validation, prospective clinical evaluation, and continuous post-deployment monitoring before routine use (NIST, 2023; U.S. Food and Drug Administration, n.d.).

Final responsibility for assessment, referral, diagnosis, and treatment should remain with qualified healthcare professionals (American Academy of Ophthalmology, n.d.; NIST, 2023; World Health Organization, 2021). Maintaining clear clinical boundaries helps ensure that AI serves as a supportive technology rather than a substitute for professional medical judgment (NIST, 2023; World Health Organization, 2021).

summary table

Table 1: Summary of Responsible AI Principles for Pediatric Vision AI-Assisted Decision Support

Principle	Why It Matters
Fairness	Equitable performance across diverse pediatric populations
Privacy	Protects sensitive health information and builds trust
Explainability	Supports clinician understanding and caregiver communication
Safety	Ensures AI augments rather than replaces clinical judgment

Note: Adapted from responsible AI frameworks (NIST, 2023; World Health Organization, 2021)

7. Validation Requirements

The successful integration of artificial intelligence into pediatric vision care requires more than strong algorithmic performance. Before AI-assisted decision support can be considered for routine clinical use, comprehensive validation is necessary to demonstrate not only technical reliability but also clinical effectiveness, patient safety, workflow compatibility, and equitable performance across diverse populations (NIST, 2023; U.S. Food and Drug Administration, n.d.; World Health Organization, 2023).

The proposed framework described in this white paper is conceptual and has not undergone clinical validation. Accordingly, future research should focus on establishing rigorous validation processes that evaluate AI systems under real-world healthcare conditions rather than relying solely on retrospective model performance (He et al., 2025; Park et al., 2025; Shu et al., 2024).

The following validation domains represent key considerations for future investigation.

7.1 Clinical Validation Needs

Technical measures such as accuracy, sensitivity, specificity, and area under the receiver operating characteristic curve (AUC) provide valuable information regarding model performance; however, these metrics alone are insufficient to establish clinical utility (U.S. Food and Drug Administration, n.d.).

This distinction is particularly relevant in pediatric vision care, where the ultimate measure of success is not model accuracy but whether children identified as potentially at risk are successfully connected to timely ophthalmologic evaluation and receive appropriate intervention within the critical developmental window (American Academy of Ophthalmology, n.d.; American Association For Pediatric Ophthalmology And Strabismus, n.d.).

Future studies should evaluate whether AI-assisted risk identification contributes to meaningful improvements in patient care, including earlier recognition of vision concerns, increased referral completion, reduced delays in ophthalmologic evaluation, and improved clinical outcomes.(Park et al., 2025; World Health Organization, 2023) Validation should also assess clinician acceptance, usability, and the impact of AI-assisted decision support on routine pediatric practice (NIST, 2023; Park et al., 2025).

Prospective clinical studies involving multidisciplinary collaboration between pediatricians, ophthalmologists, health informatics specialists, and implementation researchers will be essential before considering widespread adoption (NIST, 2023; U.S. Food and Drug Administration, n.d.; Yanovitch, 2024).

7.2 Under-3 Population Considerations

Children under the age of three represent one of the most important yet challenging populations for pediatric vision research. Clinical assessment during this developmental period is often complicated by limited cooperation, rapidly changing developmental milestones, and subtle or intermittent clinical findings (American Academy of Ophthalmology, n.d.; American Association For Pediatric Ophthalmology And Strabismus, n.d.; National Eye Institute, n.d.-b).

Because the proposed framework specifically focuses on this early developmental window, future validation studies should include representative cohorts of infants and toddlers rather than relying primarily on preschool- or school-aged populations. Particular attention should be given to age-specific clinical presentation, developmental variability, and the unique challenges associated with evaluating vision risk in very young children (American Association For Pediatric Ophthalmology And Strabismus, n.d.; Park et al., 2025).

Existing AI studies have predominantly validated performance in children aged five years and older, with notably lower sensitivity reported in younger age groups. One recent study reported sensitivity as low as 67–69% for children under five years of age, highlighting the need for dedicated validation in younger cohorts (Shu et al., 2024).

Validation across this age group is especially important because timely intervention during the earliest years of visual development may offer the greatest opportunity to improve long-term outcomes (Hashemi et al., 2018, 2019; National Eye Institute, n.d.-b).

7.3 Real-World Workflow Testing

In addition to clinical performance, future research should evaluate how AI-assisted decision support functions within routine healthcare environments. Real-world implementation involves factors that extend beyond algorithm development, including workflow integration, clinician adoption, referral coordination, electronic health record compatibility, and long-term system sustainability (NIST, 2023; Park et al., 2025; World Health Organization, 2023).

Implementation studies should examine whether AI recommendations can be incorporated into existing pediatric workflows without increasing clinician burden or disrupting routine patient care (Park et al., 2025). Outcomes of interest may include referral adherence, follow-up completion, healthcare provider acceptance, and the overall feasibility of integrating AI-assisted risk assessment into diverse clinical settings (NIST, 2023; Park et al., 2025; World Health Organization, 2023).

Ultimately, successful validation should demonstrate not only that an AI system performs accurately, but also that it contributes to safer, more effective, and more equitable pediatric

healthcare delivery (NIST, 2023; U.S. Food and Drug Administration, n.d.; World Health Organization, 2023).

8. Future Research Directions

While recent advances in artificial intelligence have demonstrated considerable promise for pediatric vision screening, substantial work remains before AI-assisted systems can be responsibly integrated into routine clinical practice (He et al., 2025; Mao et al., 2021; Murali et al., 2021; Shu et al., 2024). Future research should extend beyond algorithm development to address the broader clinical, organizational, and implementation challenges that influence real-world adoption (NIST, 2023; Park et al., 2025; World Health Organization, 2023). Building upon the gaps identified throughout this white paper, the following research directions are proposed to support the responsible translation of AI into pediatric vision care (NIST, 2023; U.S. Food and Drug Administration, n.d.).

8.1 Clinical Collaboration

Future progress in pediatric vision AI will require close collaboration among pediatricians, pediatric ophthalmologists, orthoptists, healthcare informatics specialists, implementation scientists, data scientists, and AI researchers (American Academy of Ophthalmology, n.d.; American Association For Pediatric Ophthalmology And Strabismus, n.d.; Park et al., 2025). Interdisciplinary collaboration is essential to ensure that future AI-assisted decision support systems are clinically relevant, technically robust, ethically responsible, and compatible with existing healthcare workflows (NIST, 2023; Park et al., 2025; World Health Organization, 2023).

Partnerships between academic institutions, healthcare organizations, and research networks may facilitate the development of standardized evaluation protocols, prospective validation studies, and evidence-based implementation strategies (NIST, 2023; Park et al., 2025). Such collaboration will also support continuous refinement of AI systems based on clinical feedback and evolving healthcare needs (U.S. Food and Drug Administration, n.d.; World Health Organization, 2023).

8.2 Dataset Development

The reliability and generalizability of AI systems depend heavily on the availability of diverse, representative, and ethically governed datasets (NIST, 2023; World Health Organization, 2023). Future research should prioritize the development of pediatric datasets that include children from different demographic backgrounds, healthcare settings, and stages of visual development (He et al., 2025; Park et al., 2025; Shu et al., 2024).

A bibliometric analysis of 146 published studies on AI in strabismus confirmed that the absence of large-scale publicly available training datasets represents one of the most significant structural

barriers to progress in this field, limiting the ability of research teams to benchmark, replicate, and build upon existing work (Zhou et al., 2023).

Particular attention should be given to expanding data resources for children under the age of three, where publicly available datasets remain limited despite the importance of this developmental period (Mao et al., 2021; Shu et al., 2024). Future dataset development should also emphasize data quality, standardized clinical annotation, privacy protection, and responsible governance to support reproducible and trustworthy AI research (NIST, 2023; U.S. Food and Drug Administration, n.d.; World Health Organization, 2023).

8.3 Multi-Site Studies

Most published AI studies in pediatric vision have been conducted within relatively limited clinical environments or retrospective research datasets (He et al., 2025; Mao et al., 2021; Murali et al., 2021; Shu et al., 2024). Future investigations should include prospective, multi-center studies involving children's hospitals, community pediatric practices, academic medical centers, and diverse healthcare system (NIST, 2023; Park et al., 2025; World Health Organization, 2023).

In particular, multi-site studies should prioritise inclusion of underrepresented populations, including children from low-income communities, rural healthcare settings, and ethnically diverse backgrounds, where access to specialist pediatric ophthalmology services may be most limited and where the potential public health benefit of responsible AI-assisted screening support may be greatest (Park et al., 2025; World Health Organization, 2023).

Evaluation across multiple institutions will improve understanding of model robustness, clinical generalizability, workflow compatibility, and equitable performance across different patient populations (NIST, 2023; Park et al., 2025). Multi-site collaboration may also contribute to the development of standardized evaluation frameworks that support reproducibility and facilitate comparison across future studies (NIST, 2023; U.S. Food and Drug Administration, n.d.; World Health Organization, 2023).

8.4 Implementation Research

Demonstrating strong algorithmic performance represents only one stage in the translation of AI into clinical practice (NIST, 2023; U.S. Food and Drug Administration, n.d.). Future research should investigate how AI-assisted decision support can be effectively integrated into routine pediatric healthcare workflows while maintaining clinician oversight and minimizing disruption to existing care processes (NIST, 2023; Park et al., 2025; World Health Organization, 2023).

Implementation research should evaluate clinician acceptance, caregiver engagement, referral completion, follow-up adherence, interoperability with electronic health record systems, and

organizational readiness for adoption (NIST, 2023; Park et al., 2025). These factors are likely to influence the long-term effectiveness of AI-assisted pediatric vision care as much as technical model performance (Park et al., 2025; U.S. Food and Drug Administration, n.d.; World Health Organization, 2023).

Beyond healthcare institutions, future research should also explore collaboration with public health organizations, professional societies, educators, and community stakeholders to strengthen awareness, promote evidence-based screening practices, and support equitable access to pediatric eye care (American Academy of Ophthalmology, n.d.; American Association For Pediatric Ophthalmology And Strabismus, n.d.; World Health Organization, 2023). Integrating perspectives from multiple stakeholders may improve implementation strategies and facilitate responsible adoption as supporting scientific evidence continues to evolve (NIST, 2023; Park et al., 2025; World Health Organization, 2023).

Ultimately, future research should move beyond asking whether AI can detect pediatric vision disorders toward understanding how AI can responsibly strengthen the entire continuum of care—from early risk identification and timely referral to coordinated follow-up, equitable access, and improved long-term visual outcomes (NIST, 2023; Park et al., 2025; U.S. Food and Drug Administration, n.d.; World Health Organization, 2023).

9. Limitations

This white paper presents a conceptual research framework and should be interpreted within that context. The proposed hospital-integrated approach has not been clinically validated, prospectively evaluated, or implemented within a real-world healthcare environment. Accordingly, the framework should not be interpreted as a validated clinical system, diagnostic model, or medical device (NIST, 2023; U.S. Food and Drug Administration, n.d.).

The framework is primarily intended to stimulate academic discussion and future interdisciplinary research rather than provide immediate recommendations for clinical practice. While the concepts presented are informed by the current literature on pediatric vision screening, healthcare artificial intelligence, and responsible AI, they remain theoretical and require substantial empirical investigation before consideration for clinical implementation (NIST, 2023; World Health Organization, 2023).

In addition, this white paper does not propose a specific machine learning architecture, predictive algorithm, or software implementation. Instead, it focuses on the broader integration of responsible AI within pediatric healthcare workflows, recognizing that technical design decisions will depend on future research, available clinical data, institutional requirements, and regulatory considerations (NIST, 2023; U.S. Food and Drug Administration, n.d.).

The literature reviewed for this work primarily reflects currently available published evidence. As artificial intelligence, pediatric ophthalmology, and healthcare technologies continue to evolve rapidly, future scientific discoveries, clinical guidelines, and regulatory developments may influence or refine the concepts presented in this framework (He et al., 2025; Park et al., 2025; World Health Organization, 2023).

Furthermore, the proposed workflow has not yet been evaluated across diverse healthcare systems, geographic regions, or patient populations. Future prospective studies involving multidisciplinary collaboration and multi-site clinical validation will be essential to assess its feasibility, safety, generalizability, and potential impact on pediatric vision care (NIST, 2023; Park et al., 2025; World Health Organization, 2023).

Despite these limitations, the framework aims to provide a structured foundation for future investigation into the responsible integration of artificial intelligence within pediatric vision care. By emphasizing healthcare workflows, clinician oversight, and early risk identification, this work seeks to encourage interdisciplinary collaboration and contribute to the ongoing development of evidence-based, patient-centered AI applications in pediatric ophthalmology (American Academy of Ophthalmology, n.d.; American Association For Pediatric Ophthalmology And Strabismus, n.d.; NIST, 2023; World Health Organization, 2023).

10. Conclusion

Early childhood represents a critical window for visual development, and conditions such as amblyopia and strabismus can lead to long-term visual impairment when identification and intervention are delayed. Although significant progress has been made in pediatric vision screening and artificial intelligence research, important gaps remain between technical advances in AI-based detection and the realities of clinical implementation, referral coordination, and continuity of care (He et al., 2025; Park et al., 2025; Shu et al., 2024).

This white paper has presented a conceptual hospital-integrated framework for responsible AI-assisted pediatric vision risk identification before the age of three. Rather than proposing an autonomous diagnostic system, the framework emphasizes the integration of AI-assisted decision support within routine pediatric healthcare workflows, supported by electronic health record information, clinician oversight, and structured referral pathways (NIST, 2023; World Health Organization, 2023).

The central argument of this work is that improving pediatric vision outcomes requires moving beyond isolated model performance metrics toward a broader healthcare systems perspective (Park et al., 2025; World Health Organization, 2023). Early risk identification is only meaningful if it is connected to timely ophthalmologic evaluation, effective follow-up, and coordinated care delivery (American Academy of Ophthalmology, n.d.; American Association For Pediatric

Ophthalmology And Strabismus, n.d.). Consequently, future research should focus not only on algorithm development, but also on validation in real-world clinical environments, equitable performance across diverse populations, workflow integration, and responsible governance (NIST, 2023; U.S. Food and Drug Administration, n.d.; World Health Organization, 2023) .

By synthesizing current literature, identifying implementation gaps, and proposing a workflow-oriented conceptual framework, this white paper seeks to contribute to ongoing interdisciplinary discussions among clinicians, researchers, healthcare administrators, public health organizations, and AI practitioners (Park et al., 2025; World Health Organization, 2023). The framework is intended as a foundation for future investigation rather than a validated clinical solution (NIST, 2023).

Ultimately, the long-term goal of responsible AI-assisted pediatric vision care should be to support earlier recognition of children at risk, strengthen referral and follow-up pathways, and help reduce preventable childhood vision loss through safer, more equitable, and more integrated healthcare delivery (American Academy of Ophthalmology, n.d.; American Association For Pediatric Ophthalmology And Strabismus, n.d.; National Eye Institute, n.d.-b; World Health Organization, 2023).

Every child deserves the opportunity for timely vision care during the years when intervention matters most.

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